The MAAT Structural Selection Principle

A Complementary Layer for Fundamental Physics

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Abstract

We formulate the MAAT framework as a complementary structural-selection layer for physics. The proposal does not replace dynamical laws and is not presented as a completed fundamental theory. Its purpose is narrower: to define an operational layer that ranks which dynamically admissible configurations are structurally favoured.

The theory is formulated as an effective structural action

$$S_{\text{eff}} = S_0 + F_{\text{MAAT}},$$

where S_0 describes standard dynamics and F_{MAAT} encodes structural selection across five sectors: consistency, balance, activity, connectivity, and robustness. The central contribution of this note is an operational input map: it specifies the six ingredients required to turn the MAAT principle from a symbolic formula into a calculable selection functional. These ingredients are a base dynamics, a reference measure, primitive defects, normalisation scales, sector weights, and a selection rule.

The logic is illustrated using the cosmological Critical Coherence Index (CCI), including growth connectivity and robustness margins. This cosmological example serves as a concrete benchmark of the general input map, not as evidence for a complete fundamental theory.

1 Introduction

Modern physics defines which configurations are dynamically allowed, but does not explicitly determine which of these configurations are realised in nature. This work proposes a complementary principle:

> *Physics defines what is dynamically admissible. Structural selection determines what is realised.*

The MAAT framework introduces a structural-selection layer on top of standard dynamics. This layer is not meant to replace general relativity, quantum field theory, cosmology, or statistical mechanics. Instead, it defines a ranking over admissible configurations using structural criteria that can be measured, normalised, and tested. This work should therefore be understood as a structural-selection layer that can be attached to different dynamical theories, rather than a complete standalone fundamental theory.

2 The Master Principle

The theory is defined by the effective structural action

$$S_{\text{eff}}[X] = S_0[X] + F_{\text{MAAT}}[X], \tag{1}$$

where X denotes the physical configuration under consideration. Depending on the domain, X may contain fields, metrics, matter distributions, parameters, computational states, or cosmological histories.

The two layers have different roles:

- $S_0[X]$ defines dynamical admissibility: equations of motion, constraints, symmetries, and conservation laws.
- $F_{\mathrm{MAAT}}[X]$ defines structural selection: which admissible or approximately admissible configurations are preferred.

In path-integral or ensemble language, the corresponding structural measure is

$$Z_{\mathrm{MAAT}}[J] = \int \mathrm{d}\mu_0[X] \, \exp\left(\frac{i}{\hbar} S_0[X] - \beta F_{\mathrm{MAAT}}[X] + J \cdot \mathcal{O}[X]\right). \tag{2}$$

Here μ_0 is the reference measure, β controls the strength of structural selection, and J couples to observables $\mathcal{O}[X]$. Equation (2) is the first operational form of the MAAT master formula used in this note.

MAAT Master Formula

Combining the dynamical layer, the reference measure, the structural supports, and the observable coupling gives the compact master formula:

$$\boxed{\begin{aligned} Z_{\mathrm{MAAT}}[J] &= \int \mathrm{d}\mu_0[X] \, \exp\left[\frac{i}{\hbar} S_0[X] - \beta F_{\mathrm{MAAT}}[X] + J \cdot \mathcal{O}[X]\right], \\ F_{\mathrm{MAAT}}[X] &= - \sum_{a \in \{H,B,S,V,R\}} \lambda_a \log(\epsilon + \Gamma_a[X]), \\ \Gamma_a[X] &= \frac{1}{1 + d_a[X]}. \end{aligned}} \tag{3}$$

This is the MAAT structural-selection master formula. Its meaning is deliberately layered:

- S_0 determines which configurations are dynamically admissible;
- μ_0 defines the unselected candidate ensemble;
- d_a measures structural defects in the five sectors;
- Γ_a converts defects into bounded supports;
- λ_a sets the importance of each sector;
- F_{MAAT} defines the structural selection cost;
- Z_{MAAT} defines the resulting weighted ensemble.

3 What Must Be Inserted

To turn the MAAT expression into a calculable framework, one must specify six ingredients. Without these ingredients, the formula remains only a symbolic architecture. This input map is the main contribution of the present note.

3.1 1. A Dynamical Base Theory

First, one must choose the dynamical layer S_0 and the configuration space $\mathcal{X}$:

$$X \in \mathcal{X}, \qquad S_0[X] \text{ given.} \tag{4}$$

Examples include:

- scalar field theory: $X = \phi(x)$,
- general relativity plus matter: $X = (g_{\mu\nu}, \Phi_i)$,
- cosmology: $X = (H(z), D(z), f\sigma_8(z), \Omega_i, \ldots)$,
- string compactifications: $X = (g_{10}, F_p, \tau, \mathcal{M}_6, \text{branes}, \ldots)$,
- computational systems: $X =$ constraint instances, solution states, or algorithmic trajectories.

This step answers the question:

What counts as a possible configuration?

3.2 2. A Reference Measure

Second, one must define the unselected reference measure μ_0:

$$\mathrm{d}\mu_0[X]. \tag{5}$$

This may be a lattice field measure, a stochastic stationary measure, a prior over model parameters, an ensemble of compactifications, or an empirical data distribution. The reference measure is important because MAAT selection does not rank "nothing". It ranks configurations relative to a prior space of candidates.

For example, in a lattice ϕ^4 benchmark, μ_0 may be the measure over lattice field values and momenta, or in a stochastic formulation the stationary measure of the uncontrolled Langevin dynamics. In cosmology, μ_0 may be a reference ΛCDM model plus observational data tables.

3.3 3. Primitive Defects

Third, one must define primitive defect functionals

$$d_a[X] \geq 0, \qquad a \in \{H, B, S, V, R\}. \tag{6}$$

These defects measure how strongly a configuration fails each structural criterion. A minimal canonical choice is:

$$d_H[X] \sim \mathcal{N}_H \left[\left\| \frac{\delta S_0}{\delta X} \right\|^2 \right], \qquad \text{consistency,} \tag{7}$$

$$d_B[X] \sim \mathcal{N}_B \left[\|\nabla_\mu T^{\mu\nu}\|^2 + \|\mathcal{C}_{\text{constraints}}\|^2 \right], \qquad \text{balance,} \tag{8}$$

$$d_S[X] \sim \mathcal{N}_S[\text{activity}], \qquad \text{production,} \tag{9}$$

$$d_V[X] \sim \mathcal{N}_V[\text{correlation}], \qquad \text{connectivity,} \tag{10}$$

$$d_R[X] \sim \mathcal{N}_R[\text{stability margin}], \qquad \text{robustness.} \tag{11}$$

The maps $\mathcal{N}_a$ are normalisations that make the defects dimensionless. In words: H measures equation residuals, B measures conservation and constraint residuals, S measures controlled dynamical activity, V measures correlation or coupling support, and R measures stability margins under perturbations or constraint pressure.

3.4 4. Normalisation and Cutoffs

Fourth, one must specify the scales or cutoffs that make each defect dimensionless and comparable:

$$d_a[X] = \frac{\mathcal{D}_a[X]}{\Lambda_a^{p_a} + \epsilon}. \tag{12}$$

The scale Λ_a may be a physical cutoff, a lattice scale, an observational uncertainty, a reference-model scale, or a data-driven normalisation. This step is essential. Without it, one sector can dominate merely because it is measured in larger units.

The bounded support form is

$$\Gamma_a[X] = \frac{1}{1 + d_a[X]}, \qquad 0 < \Gamma_a \leq 1. \tag{13}$$

High support means low defect.

3.5 5. Sector Weights

Fifth, one must determine the weights λ_a:

$$\lambda_a > 0. \tag{14}$$

These weights may be chosen in three increasingly strong ways:

- unit convention: $\lambda_a = 1$ for a first unfitted diagnostic;
- maximum-entropy calibration: infer λ_a from ensemble constraints;
- domain theory: derive λ_a from symmetries, scales, or microscopic physics.

The present research programme has already tested the first two levels in field, optimisation, effective-constant, and cosmological benchmarks. The third level remains the deeper open problem.

3.6 6. A Selection Rule

Finally, one must define the structural functional and the selection rule. The nonlinear support form is

$$\boxed{F_{\mathrm{MAAT}}[X] = -\sum_{a \in \{H,B,S,V,R\}} \lambda_a \log(\epsilon + \Gamma_a[X])} \tag{15}$$

or, for small-defect calibration and weight-fitting benchmarks, the linear proxy

$$F_{\mathrm{MAAT}}[X] \approx \sum_a \lambda_a d_a[X]. \tag{16}$$

This distinction is important for consistency with earlier MAAT calibration papers. The nonlinear logarithmic form is the canonical support-based functional, because it treats each sector as a multiplicative support factor in the selection probability. The linear form is a practical proxy used when fitting sector weights in closed benchmark datasets, such as the boundary-aware calibration study. The selection probability is

$$P[X] = \frac{1}{Z} \exp[-F_{\mathrm{MAAT}}[X]]. \tag{17}$$

The preferred configuration is then

$$X_{\mathrm{selected}} = \arg\min_{X \in \mathcal{X}_{\mathrm{adm}}} F_{\mathrm{MAAT}}[X], \tag{18}$$

where $\mathcal{X}_{\mathrm{adm}}$ is the dynamically admissible region defined by S_0.

4 Structural Selection Functional

In local field-theory form, the structural functional may be written as

$$F_{\mathrm{MAAT}}[X] = \int \mathrm{d}^d x \, \sqrt{|g|} \sum_{a \in \{H,B,S,V,R\}} \lambda_a \rho_a(x; X), \tag{19}$$

where $\rho_a(x; X)$ are local structural densities. The local form is useful for field theory and gravity, while the support form in Eq. (15) is useful for ranking, probability, and cross-domain comparison.

5 The Five Structural Sectors

The five sectors can be read in plain language as follows:

- H – consistency: does the configuration satisfy its equations?
- B – balance: are constraints, symmetries, and conservation laws respected?
- S – activity: does the system contain controlled dynamical richness rather than trivial inactivity or destructive excitation?
- V – connectivity: are the degrees of freedom coherently coupled, correlated, or integrated?
- R – robustness: does the configuration remain stable under perturbations and near constraints?

Thus MAAT does not add five new forces. It adds five diagnostic sectors that measure the structural quality of configurations already defined by the base theory.

6 Connection to the Cosmological CCI

The cosmological Critical Coherence Index provides one concrete benchmark for the structural-selection logic. In the broader CCI paper series, the CCI was introduced as a reduced observable measuring the balance between activity and coherence. In the cosmological version, the same logic becomes:

$$\mathrm{CCI}_{\cos}(z) = \frac{E(z)(1+z)\left[1 + U_{\mathrm{struct}}(z)\right]}{1 + D(z) + V(z) + R(z) + \epsilon}. \tag{20}$$

This combines:

- expansion history $H(z)$ through $E(z) = H(z)/H_0$,
- growth factor $D(z)$,
- connectivity from $f\sigma_8(z)$,
- robustness from joint expansion–growth consistency.

In the v0.3 cosmology benchmark, the operational substitutions are:

$$d_H(z) = \text{local chronometer } H(z) \text{ residual}, \tag{21}$$

$$d_S(z) = \frac{|E(z) - D(z)|}{E(z) + D(z) + \epsilon}, \tag{22}$$

$$d_V(z) = \text{local } f\sigma_8(z) \text{ residual}, \tag{23}$$

$$d_R(z) = \frac{1}{2}\left[d_H(z) + d_V(z)\right]. \tag{24}$$

This shows explicitly what must be inserted into the master formula to make the cosmological sector calculable. It should not be read as carrying the full weight of the MAAT framework. Rather, it demonstrates how the general input map can be instantiated in one domain using measured expansion and growth observables.

7 Key Result

The cosmological CCI reveals:

- a structural-stress history of the universe;
- a transition marker at $z_c \approx 1.1$ in the v0.3 benchmark;
- measurable multi-sector behaviour through H, S, V, and R;
- effective maximum-entropy sector weights in the tested dataset.

8 Interpretation

While $H(z)$ describes expansion, the CCI describes structural imbalance.

The universe evolves from a high-stress regime toward a coherence-supported late-time structure. In MAAT language, cosmic evolution is not only a history of expansion, but a history of the changing balance between activity, connectivity, and robustness.

9 Relation to Other Fundamental-Physics Programmes

The MAAT proposal is best understood as a selection layer rather than a replacement dynamics. This makes it complementary to several existing approaches to unification.

Table 1: Conceptual comparison between the MAAT structural-selection layer and other fundamental-physics programmes.

Programme	**Primary aim**	**MAAT relation**
Standard Model + GR	Describes known particle interactions and classical gravitation.	MAAT does not replace these laws. It adds a structural ranking over their admissible configurations.
String theory	Provides a candidate quantum-gravity framework with many possible compactifications and vacua.	MAAT can act as a pre-anthropic selection measure over the string landscape, ranking admissible backgrounds by structural support.
Loop quantum gravity / spin foams	Quantises geometry and spacetime histories without assuming a fixed background.	MAAT could rank spin-foam or quantum-geometry histories by consistency, connectivity, and robustness, but does not derive discreteness itself.
Asymptotic safety	Searches for a UV-complete gravitational fixed point.	MAAT is complementary: it can evaluate structural stability of flows or effective theories, but does not by itself prove a UV fixed point.
Causal dynamical triangulations / causal sets	Build spacetime from causal or triangulated microscopic structures.	MAAT can be interpreted as a structural weighting over causal histories, but the microscopic causal construction remains part of the base theory.
Hartle–Hawking / no-boundary measures	Define cosmological wavefunction or boundary-condition weights.	MAAT differs by using multi-sector structural supports rather than only a cosmological boundary prescription.
Anthropic / landscape measures	Select universes by observer-compatible conditions or environmental constraints.	MAAT is intended as a pre-anthropic structural prior: it ranks backgrounds before imposing observer selection.

The central distinction is therefore:

> *Most fundamental-physics programmes try to define the underlying dynamics. MAAT asks which dynamically admissible configurations are structurally selected.*

This is why the framework can be attached to different base theories. If S_0 is string theory, MAAT becomes a landscape-selection functional. If S_0 is quantum gravity, MAAT becomes a weighting over quantum-geometric histories. If S_0 is effective cosmology, MAAT becomes a structural-stress diagnostic over cosmic histories.

10 What This Is and What It Is Not

This is:

- a structural-selection principle;
- a testable extension of physics;
- a multi-scale framework;
- an operational recipe for constructing selection functionals.

This is not:

- a replacement for standard physics;
- a complete final theory;
- a parameter-fitting cosmological model;
- a proof that all constants, particles, fields, or interactions have already been derived.

11 Figure

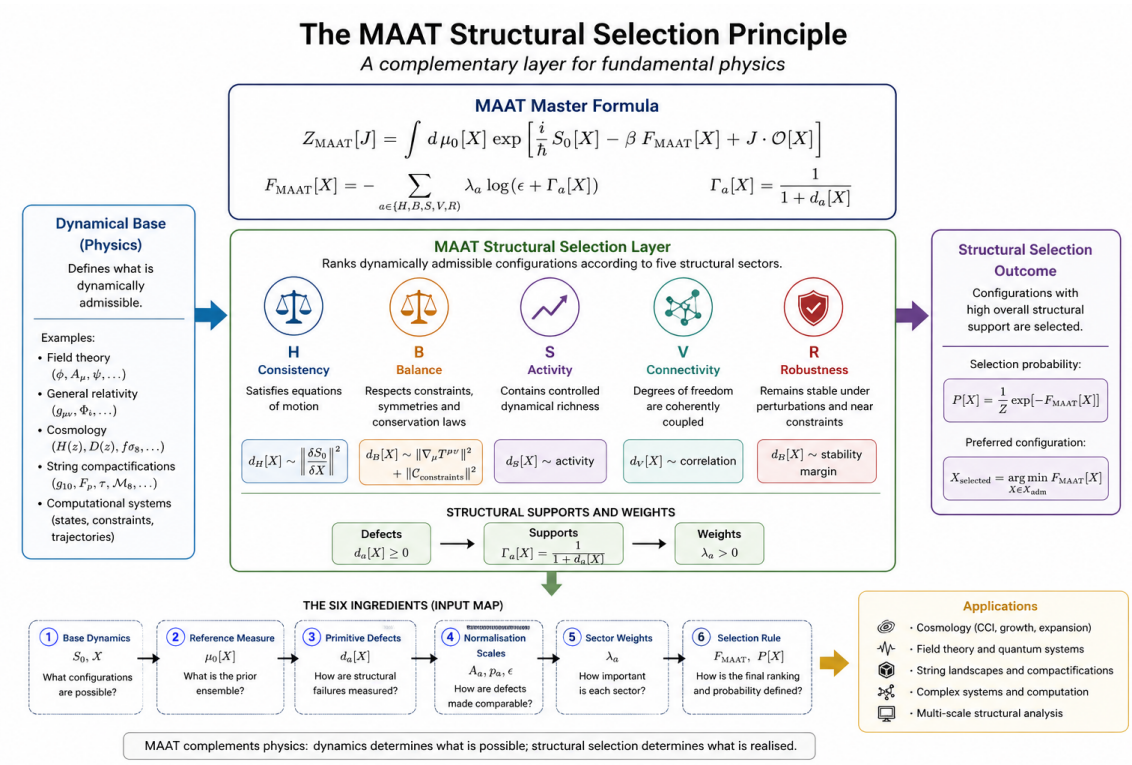

Figure 1: MAAT structural-selection overview: structural selection is proposed as a candidate layer on top of dynamical physics, not as a replacement for the underlying dynamics.

12 Outlook

Future work includes:

- integration of BAO, DESI, supernova, and weak-lensing data;
- full perturbation-theory implementation;
- application to field theory, compactifications, and computation;

- derivation of sector weights from deeper microscopic principles;
- extension toward a complete MAAT functional with explicit reference measures, cutoffs, and falsification tests in each domain.

13 Conclusion

We formulated MAAT as a structural-selection layer that can be attached to dynamical theories without replacing them. The framework provides an operational way to compare field-theoretic, cosmological, and complex-system configurations through common structural diagnostics.

The key clarification is that the master formula becomes meaningful only after six ingredients are specified: a dynamical base theory, a reference measure, primitive defects, normalisation scales, sector weights, and a selection rule. Once these are inserted, MAAT becomes a calculable structural-selection architecture rather than a purely symbolic proposal.